

Ch 5: Life Processes

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What are life processes? Name the four essential life processes studied in this chapter. (PYQ 2019)

Q2. What is nutrition? What are the two main types of nutrition? Give one example of each. (PYQ 2020)

Q3. What is autotrophic nutrition? What are the raw materials required by plants for photosynthesis? (PYQ 2022)

Q4. Write the balanced chemical equation for photosynthesis. Name the site in the cell where photosynthesis takes place. (PYQ 2019)

Q5. What is the role of chlorophyll in photosynthesis? What happens to photosynthesis at night? (PYQ 2022)

Q6. What are stomata? State two functions of stomata. What causes the stomata to open and close? (PYQ 2020)

Q7. What is heterotrophic nutrition? Name three types of heterotrophic nutrition with one example each. (PYQ 2019)

Q8. What is the function of saliva in digestion? Name the enzyme present in saliva and state its function. (PYQ 2022)

Q9. What is the role of the stomach in digestion? Name the acid secreted by the stomach and state its functions. (PYQ 2020)

Q10. What is the role of the small intestine in digestion? Name the enzymes secreted here and the substance each acts on. (PYQ 2019)

Q11. What is the role of bile in digestion? Where is bile produced? Where is it stored? What is emulsification? (PYQ 2022)

Q12. What is the role of villi in the small intestine? How do nutrients reach all cells of the body from the intestine? (PYQ 2020)

Q13. What is respiration? How is respiration different from breathing? (PYQ 2019)

Q14. Write the balanced equation for aerobic respiration. Where does it take place in the cell? (PYQ 2022)

Q15. Write the equation for anaerobic respiration in: (i) yeast (ii) human muscle cells during vigorous exercise. (PYQ 2020)

Q16. Why do athletes suffer from muscle cramps after vigorous exercise? What is the substance that accumulates? (PYQ 2024)

Q17. How do fish breathe in water? What is the function of gills? (PYQ 2019)

Q18. What is the role of haemoglobin in respiration and transportation of gases? (PYQ 2022)

Q19. What is double circulation? Why is it necessary in humans? (PYQ 2020)

Q20. What is the difference between arteries and veins? Give two differences. (PYQ 2019)

Q21. What are platelets? What is their function? What is blood pressure? (PYQ 2022)

Q22. What is lymph? How is it different from blood? What is its function? (PYQ 2020)

Q23. How does water enter the root hair cells of plants? Name the process. What is osmosis? (PYQ 2019)

Q24. What is transpiration? State two functions of transpiration. What is transpiration pull? (PYQ 2022)

Q25. What is translocation? Through which tissue does it occur? In which direction does translocation occur? (PYQ 2024)

Q26. What is excretion? Name the main excretory organ in humans. Name the functional unit of the kidney. (PYQ 2020)

Q27. What is urine? Name the main nitrogenous waste product excreted in humans. What is the composition of urine? (PYQ 2019)

Q28. How do plants excrete their waste products? Name any two excretory products of plants. (PYQ 2022)

Q29. What is dialysis? When is it required? What does a dialysis machine do? (PYQ 2020)

Q30. Name the excretory products and excretory organs in the following organisms: (i) earthworm (ii) insects (iii) fish. (PYQ 2019)

Q31. What is the process of transport in plants called when it occurs through the phloem? How is it different from the transport through xylem? (PYQ 2022)

Q32. What are the components of blood? State one function of each component. (PYQ 2020)

SECTION B - 3 MARKS

Q1. What is photosynthesis? Write the balanced equation. Explain the two stages of photosynthesis and where each stage takes place. (PYQ 2019)

Q2. Compare autotrophic and heterotrophic nutrition: (PYQ 2020)

Basis of Comparison	Autotrophic Nutrition	Heterotrophic Nutrition
Source of food		
Raw materials required		
Type of organisms		
Example		

Q3. Describe the process of digestion in the human alimentary canal. Trace food from mouth to small intestine and state what happens to it at each stage. (PYQ 2022)

Q4. Compare aerobic and anaerobic respiration: (PYQ 2019)

Basis of Comparison	Aerobic Respiration	Anaerobic Respiration
Presence of oxygen		
Site in cell		
End products		
Amount of energy released		
Chemical equation		

Q5. Explain the mechanism of breathing in humans. What are the roles of the diaphragm and intercostal muscles? State the difference between breathing and respiration. (PYQ 2022)

Q6. Compare the transport systems in plants and animals: (PYQ 2020)

Basis	Transport in Plants	Transport in Animals
Fluid used		

Basis	Transport in Plants	Transport in Animals
Tissue / system involved		
Direction of flow		
Presence of pump		
Substances transported		

Q7. Write the differences between xylem and phloem: (PYQ 2019)

Basis	Xylem	Phloem
Function		
Type of cells		
Direction of transport		
Substances transported		
Process driving transport		

Q8. Explain the structure and function of the human heart. What is the role of the four chambers of the heart? (PYQ 2022)

Q9. Describe the pathway of blood in the human circulatory system. Distinguish between pulmonary circulation and systemic circulation. (PYQ 2020)

Q10. Compare arteries, veins, and capillaries: (PYQ 2019)

Basis	Arteries	Veins	Capillaries
Direction of blood flow			
Wall thickness			
Valves present?			
Type of blood carried			
Function			

Q11. What is blood pressure? What are systolic and diastolic pressure? What is the normal blood pressure in humans? What is hypertension? (PYQ 2022)

Q12. Explain the process of urine formation in the nephron. Name the three steps involved. What happens to useful substances? (PYQ 2020)

Q13. What is the structure of a nephron? Draw a labelled diagram and explain the function of each part: glomerulus, Bowman's capsule, tubule, and collecting duct. (PYQ 2019)

Q14. Compare the modes of excretion in different organisms: (PYQ 2022)

Organism	Excretory Product	Excretory Organ	Mode of Excretion
Humans			
Fish			
Insects			
Earthworms			
Plants			

Q15. Describe how water and minerals are absorbed by the roots and transported to the leaves. What is the role of osmosis and active transport in root absorption? (PYQ 2020)

Q16. What is the significance of transpiration? How does water move from roots to leaves through xylem? Explain the cohesion-tension theory briefly. (PYQ 2019)

Q17. Explain how food (sucrose) is transported in plants through phloem. What is the source and sink in translocation? (PYQ 2022)

Q18. What are the different modes of nutrition in animals? Explain parasitic nutrition and saprophytic nutrition with one example each. (PYQ 2020)

Q19. Describe the nutrition in Amoeba. How does Amoeba capture its food? Name the process by which Amoeba digests its food. (PYQ 2019)

Q20. Draw a labelled diagram of the human digestive system and mark the following: mouth, oesophagus, stomach, liver, small intestine, large intestine. State

the function of each organ. (PYQ 2022)

Q21. Explain the role of the following in human digestion: (PYQ 2020)

Organ / Secretion	What It Secretes	Function in Digestion
Salivary glands		
Stomach wall		
Liver		
Pancreas		
Small intestine wall		

Q22. How does respiration occur in plants? Do plants have a specialised respiratory system? How do gases exchange in roots, stems, and leaves? (PYQ 2019)

Q23. Describe the respiratory system in humans. Draw a labelled diagram of the human respiratory system and explain the path of air from nose to alveoli. (PYQ 2022)

Q24. Explain the structure of the alveolus. Why are alveoli suited for gaseous exchange? What is the advantage of large surface area in lungs? (PYQ 2020)

Q25. What are the components of blood and their functions: (PYQ 2019)

Component of Blood	Where Produced	Structure / Feature	Function
Red Blood Cells (RBCs)			
White Blood Cells (WBCs)			
Platelets			
Plasma			

Q26. What is osmoregulation? What role do kidneys play in maintaining water balance in the body? (PYQ 2022)

Q27. What is dialysis? Draw a labelled diagram of a dialysis machine. When is kidney dialysis necessary? (PYQ 2020)

Q28. Explain how plants excrete their waste products. Name the organs / structures involved in plant excretion. How is excretion in plants different from that in animals? (PYQ 2019)

Q29. Write the differences between the following: (PYQ 2022)

Basis	Breathing	Respiration
Definition		
Process type (physical / chemical)		
Site		
Energy released?		

Basis	Aerobic Respiration	Anaerobic Respiration
Oxygen required?		
End products		
Energy released (ATP)		
Occurs in		

Q30. What happens when: (i) the rate of photosynthesis equals the rate of respiration in a plant (ii) a person runs very fast (iii) dialysis of blood is stopped in a kidney patient? (PYQ 2020)

SECTION C - 5 MARKS

Q1. Describe the process of photosynthesis in detail. Write the balanced equation. Explain the two stages — light reaction and dark reaction — and where each takes place. State the factors affecting the rate of photosynthesis. (PYQ 2022)

Q2. Draw a labelled diagram of the human alimentary canal. Explain the process of digestion that occurs in the: (i) mouth (ii) stomach (iii) small intestine. Name the enzymes involved at each stage and the substrates they act on. (PYQ 2020)

Q3. Compare nutrition in humans and plants across five parameters. Then describe any one mode of heterotrophic nutrition (parasitic / saprophytic / holozoic) in detail with an example: (PYQ 2019)

Parameter	Nutrition in Humans	Nutrition in Plants
Type of nutrition		
Source of food / energy		
Organ of ingestion		
Process of digestion		
Mode of energy release		

Q4. Describe the human respiratory system with a labelled diagram. Explain the mechanism of inspiration and expiration. How does gaseous exchange take place at the alveoli level? (PYQ 2022)

Q5. Compare aerobic and anaerobic respiration in detail. Write the chemical equations for both. Give one example of anaerobic respiration occurring in an organism and one in human cells. Explain why aerobic respiration releases more energy. (PYQ 2020)

Q6. Draw a labelled diagram of the human heart. Explain the working of the heart — how blood moves through the four chambers, the valves, and the two circuits of circulation (pulmonary and systemic). What is double circulation and why is it necessary? (PYQ 2019)

Q7. Describe the transport of water and minerals in plants. Explain: (i) how water is absorbed by root hair cells (ii) how it moves up through xylem (iii) the role of transpiration pull. Also explain the transport of food through phloem (translocation). (PYQ 2022)

Q8. Compare the transport systems in plants and animals across five parameters. Draw a labelled diagram of the cross-section of a dicot stem showing xylem and phloem: (PYQ 2020)

Parameter	Transport in Plants	Transport in Animals
Transporting fluid		
Tissue or system used		
Driving force / pump		
Direction of transport		
Substances carried		

Q9. Draw a labelled diagram of a nephron. Explain urine formation in three steps: glomerular filtration, tubular reabsorption, and tubular secretion. How does the kidney maintain water balance? What happens during dialysis? (PYQ 2019)

Q10. Describe the excretory system in humans. Draw a labelled diagram of the human excretory system. Explain the pathway of urine from formation to excretion. Name the components of urine. (PYQ 2022)

Q11. Describe the nutrition in Amoeba. Draw a labelled diagram showing the process of ingestion in Amoeba. Explain how Amoeba engulfs, digests, and absorbs food. What is the role of the food vacuole? (PYQ 2020)

Q12. Answer the following questions related to life processes: (PYQ 2023)

Question	Answer
What is the compensation point in plants?	
Why do desert plants keep their stomata closed during the day?	
What is the role of mucus in the stomach?	

Question	Answer
Why is the small intestine longer than the large intestine?	
What would happen if there were no platelets in the blood?	

Q13. Describe the respiratory system in a fish. How do fish breathe? How is respiration in fish different from that in humans? Draw a labelled diagram of the gills of a fish and explain their role in gaseous exchange. (PYQ 2019)

Q14. Case Study: A patient is suffering from kidney failure. His kidneys cannot filter blood properly. (PYQ 2024 case-based)

Question	Answer
What is the function of the kidneys in a healthy person?	
Name the functional unit of the kidney and describe its structure.	
What waste products accumulate in the blood due to kidney failure?	
What is dialysis? How does it mimic kidney function?	
What is kidney transplant? Is it a permanent solution?	

Q15. Write a comprehensive note on the life process of nutrition covering: (i) definition and importance (ii) autotrophic nutrition with photosynthesis equation (iii) heterotrophic nutrition with three types and examples (iv) nutrition in Amoeba (v) digestion in humans — enzyme table. (PYQ 2022 comprehensive)

Site of Digestion	Secretion	Enzyme(s)	Substrate	Product
Mouth				
Stomach				
Small intestine (pancreatic juice)				
Small intestine (intestinal juice)				

Site of Digestion	Secretion	Enzyme(s)	Substrate	Product
Small intestine (bile)				